

22/01/25

* Cartesian Product:- Let A and B are two sets. The Cartesian product of A and B, denoted by $A \times B$, is defined as follows:-

$$[A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\},]$$

$A \times B$ is the set of all possible ordered pairs whose first component comes from A and whose second component comes from B.

Example:- Let $A = \{1, 2, 3\}$
 $B = \{4, 5\}$

then

$$A \times B = \{(1, 4), (1, 5), (2, 4), (2, 5), (3, 4), (3, 5)\}$$

Note:- The Cartesian Product, also known as the 'Cross product'.

* Properties of Cartesian Product

- (i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$
- (ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$
- (iii) $A \times (B - C) = (A \times B) - (A \times C)$
- (iv) $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$

(v) If $A \subset B$, then $(A \times C) \subset (B \times C)$

(vi) $A \times B \neq B \times A$

(vii) $(A \times B) \times C \neq A \times (B \times C)$

* Cardinality of Set

The cardinality of a set is defined as the number of elements in a set. It can be finite or infinite.

Example:- The cardinality of set $A = \{1, 2, 3, 4, 5, 6\}$ is equal to 6

Because set A has six elements.

The cardinality of a set is also known as the size of set. It is denoted by modulus sign on both sides of the set name. $|A| = 6$